Cambridge International AS & A Level Physics (9702) — Paper 4 Style

Synoptic Structured Question: Laser Range Finder and Target Designator

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Short description of the device: A military laser range finder and target designator emits short infrared laser pulses towards a target. The same optical head receives reflected light and determines distance using the time-of-flight method. The instrument can also deliver a coded train of pulses to "paint" a target for guided munitions. Reliable operation requires ideas from waves, quantum physics, electricity, materials, mechanics and data handling.

Useful data:

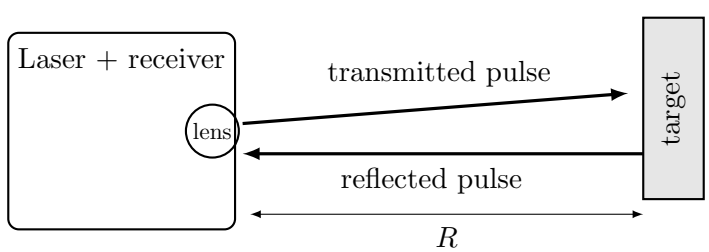
- Speed of light in vacuum: $c = 3.00 \times 10^{8}\,\mathrm{m\,s^{-1}}$
- Refractive index of glass in optics module: $n = 1.50$
- Planck constant: $h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$
- $1\ \mathrm{eV} = 1.60 \times 10^{-19}\,\mathrm{J}$

Question

Comprehension passage (read carefully before answering):
An operator uses a handheld laser range finder that emits infrared pulses of wavelength 1.064 µm. The laser emits 10 000 pulses each second, and each pulse lasts 20 ns. During each pulse, the optical output reaches a peak of 2.0 kW. In one trial, the electronics record a delay of 40.0 µs between launch of the pulse and detection of the echo from the target.

A laser range finder emits pulses of wavelength $\lambda = 1.064 \times 10^{-6}\,\mathrm{m}$. The pulse repetition frequency is 10 kHz and pulse duration is 20 ns. Peak optical power during each pulse is 2.0 kW.

Fig. 1: Simplified system layout



For one measurement, the measured time between emission and detection is $\Delta t = 40.0\,\mu\mathrm{s}$.

(a) Calculate the target distance R. [2]

(b) Explain why there is a factor of 2 in the expression for R in terms of Δt. [1]

(c) Determine the energy in one laser pulse and the average output power of the laser over many pulses. [3]

(d) Show that the maximum unambiguous range for this pulse repetition frequency is 15 km. [2]

(e) The beam divergence is 0.50 mrad. Estimate the beam diameter at a target range of 8.0 km. State one implication for target identification. [3]

(f) The optics include a glass plate of thickness 6.0 mm in the transmitted/received path.

i. Calculate the additional time delay due to this plate for one pass. [2]

ii. State whether this delay is significant compared with the pulse duration of 20 ns. [1]

(g) A photodiode receives a reflected pulse energy of 2.0×10^{-9} J at wavelength 1.064 µm. Assume each detected photon produces one electron in the diode.

i. Calculate the energy of one photon. [2]

ii. Estimate the number of electrons released. [2]

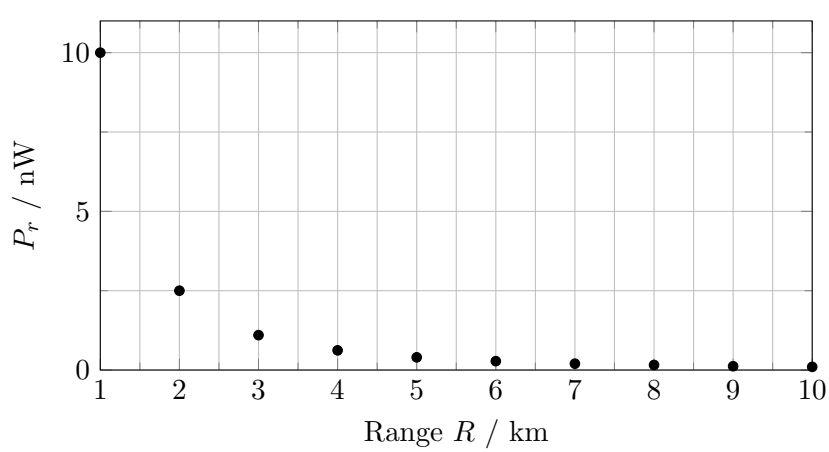
(h) The target may move directly towards the designator at $40\,\mathrm{m\,s^{-1}}$.

Determine the change in measured range between two successive pulses. Comment on whether this affects ranging precision. [3]

(i) During target designation, a coded pulse train is used so only friendly seekers lock to the intended target.

Explain, with physics reasoning, why narrow divergence, pulse coding and timing discrimination all improve immunity to background light and decoy reflections. [4]

(j) Fig. 2 shows calibration data for received optical power P_r versus range R for a fixed target reflectivity.



i. Use data from the graph to test whether $P_r \propto 1/R^2$ is consistent. [3]

ii. Suggest one physical reason why experimental points may deviate from the ideal inverse-square law at long range. [1]

[Total: 29 marks]

Suggested answers and mark scheme

General marking note: Credit correct physics methods and reasoning. Accept answers to sensible significant figures.

(a)

$$R = \frac{c\Delta t}{2} = \frac{(3.00 \times 10^8)(40.0 \times 10^{-6})}{2} = 6.00 \times 10^3\,\mathrm{m}$$

So $R = 6.0\,\mathrm{km}$.
Marks: 1 for correct formula, 1 for correct value/unit.

(b) Pulse travels to target and back, so measured time is for round trip distance $2R$.
Marks: 1 for round-trip explanation.

(c) Energy per pulse:

$$E_p = P_{\mathrm{peak}}\tau = (2.0 \times 10^3)(20 \times 10^{-9}) = 4.0 \times 10^{-5}\,\mathrm{J}$$

Average power:

$$P_{\mathrm{avg}} = E_p f = (4.0 \times 10^{-5})(1.0 \times 10^4) = 4.0 \times 10^{-1}\,\mathrm{W}$$

$P_{\mathrm{avg}} = 0.40\,\mathrm{W}$.
Marks: 1 for E_p, 1 for method for P_{avg}, 1 for value/unit.

(d) Period between pulses:

$$T = \frac{1}{f} = \frac{1}{1.0 \times 10^4} = 1.0 \times 10^{-4}\,\mathrm{s}$$

For unambiguous return, round-trip time $\leq T$:

$$R_{\mathrm{max}} = \frac{cT}{2} = \frac{(3.00 \times 10^8)(1.0 \times 10^{-4})}{2} = 1.5 \times 10^4\,\mathrm{m}$$

So $R_{\mathrm{max}} = 15\,\mathrm{km}$.
Marks: 1 for using pulse period criterion, 1 for correct value.

(e) Beam diameter $d \approx \theta R = (0.50 \times 10^{-3})(8.0 \times 10^3) = 4.0\,\mathrm{m}$.
Implication (any one): spot illuminates several objects / reduces ability to isolate one small target / increases chance of wrong reflection.
Marks: 2 for numerical estimate, 1 for valid implication.

(f) i. Additional delay for one pass through plate:

$$\Delta t_g = \frac{(n-1)L}{c} = \frac{(1.50-1)(6.0 \times 10^{-3})}{3.00 \times 10^8} = 1.0 \times 10^{-11}\,\mathrm{s}$$

(about 10 ps).
Marks: 1 for method, 1 for value.

ii. 10 ps is much smaller than 20 ns (factor ~ 2000), so negligible in this context.
Marks: 1 for correct comparison/conclusion.

(g) i. Photon energy:

$$E_\gamma = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{1.064 \times 10^{-6}} \approx 1.87 \times 10^{-19}\,\mathrm{J}$$

Marks: 1 for formula, 1 for value.

ii.

$$N = \frac{E_{\text{pulse}}}{E_\gamma} = \frac{2.0 \times 10^{-9}}{1.87 \times 10^{-19}} \approx 1.1 \times 10^{10}$$

Number of electrons $\approx 1.1 \times 10^{10}$.
Marks: 1 for method, 1 for value.

(h) Time between pulses $= T = 1.0 \times 10^{-4}\,\mathrm{s}$. Range change per pulse:

$$\Delta R = vT = (40)(1.0 \times 10^{-4}) = 4.0 \times 10^{-3}\,\mathrm{m} = 4\,\mathrm{mm}$$

This is very small; usually below metre-scale field ranging uncertainty, so minimal effect on precision for this speed.
Marks: 2 for correct calculation, 1 for sensible comment.

(i) Expected points include:

- Narrow divergence concentrates intensity on intended line of sight, reducing illumination of nearby clutter.
- Pulse coding (specific temporal pattern) lets receiver correlate only matching sequences, rejecting random/continuous background.
- Timing discrimination (range gate) accepts photons only in expected arrival window, suppressing off-range reflections.
- Combined effect improves signal-to-noise ratio and reduces false lock from decoys/sunlight.

Marks: any 4 valid, physics-linked points.

(j) i. Test using $P_r R^2$:
At $R = 2\,\mathrm{km}$, $P_r R^2 = 2.5 \times 4 = 10$ (nW km^2)
At $R = 6\,\mathrm{km}$, $P_r R^2 = 0.28 \times 36 = 10.08$ (nW km^2)
At $R = 10\,\mathrm{km}$, $P_r R^2 = 0.10 \times 100 = 10$ (nW km^2)
Approximately constant, so data are consistent with $P_r \propto 1/R^2$ within scatter.
Marks: 2 for valid numerical check, 1 for conclusion.

ii. Any one: atmospheric absorption/scattering, turbulence-induced beam spread, target orientation changes, detector noise floor/background becoming significant.
Marks: 1 for valid reason.

Variant 2 (Higher difficulty): Laser Range Finder and Coded Target Designation in Adverse Conditions

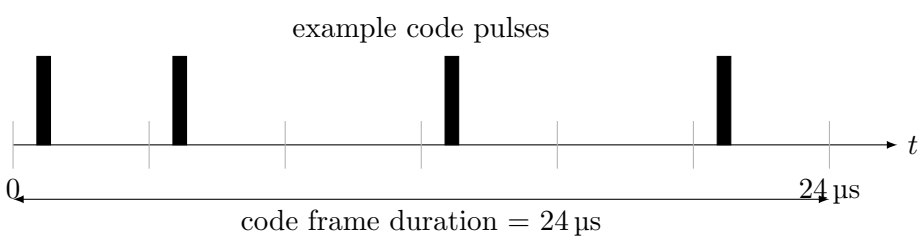
Comprehension passage (read carefully before answering):
A stabilised designator operates at wavelength 1.55 µm. It emits short pulses of duration 8.0 ns at a repetition frequency of 50 kHz, with peak power 1.6 kW. The receiver uses a 60 mm aperture lens and a comparator threshold of 18 mV. In one measurement frame, the observed round-trip delay is 73.6 µs, and beam-pointing jitter has standard deviation 30 µrad.

Short description of the device: An advanced laser target designator is mounted on a stabilised platform. It measures range by time-of-flight and transmits an identification code using pulse-position modulation. In haze and thermal shimmer, the returning signal is weak and spread in time, so the electronics use amplification, thresholding and range-gating to recover the true target return.

Additional data for Variant 2:

- Laser wavelength $\lambda = 1.55\,\mu\text{m}$
- Pulse repetition frequency $f = 50\,\text{kHz}$
- Pulse width $\tau = 8.0\,\text{ns}$
- Peak optical power $P_{\text{peak}} = 1.6\,\text{kW}$
- Receiver lens diameter $D = 60\,\text{mm}$
- Angular jitter (1σ) of outgoing beam: $\sigma_\theta = 30\,\mu\text{rad}$
- Comparator threshold in receiver: $V_T = 18\,\text{mV}$

Fig. 3: Timing and coding frame



At one instant, the measured round-trip delay is $\Delta t = 73.6\,\mu\text{s}$ to a moving vehicle. The vehicle is known to move directly away from the instrument at approximately constant speed.

(a) Calculate the measured range at this instant. [2]

(b) The timing electronics have uncertainty of $\pm 2.0\,\text{ns}$ in locating pulse arrival.

i. Determine the corresponding uncertainty in range measurement. [2]

ii. Give one design change to reduce this timing-based uncertainty. [1]

(c) i. Calculate pulse energy and average transmitted power. [3]

ii. Calculate the number of photons emitted per pulse. [2]

(d) The round-trip atmospheric attenuation is represented by a transmission factor

$$T(R) = \exp(-2\alpha R), \quad \alpha = 1.8 \times 10^{-4}\,\mathrm{m}^{-1}.$$

For $R = 4.0\,\mathrm{km}$, calculate T and hence the received pulse energy if the overall geometric/reflectivity factor is 3.0×10^{-8} (before atmospheric attenuation). [4]

(e) The receiver transimpedance amplifier output for a detected pulse is approximated by

$$V(t) = V_0 e^{-t/\tau_r}, \quad \tau_r = 12\,\mathrm{ns},$$

where $t = 0$ at pulse peak and $V_0 = 42\,\mathrm{mV}$.

i. Determine the time interval after the peak for which $V(t) > V_T$. [2]

ii. Explain why excessive threshold voltage may cause systematic range error. [2]

(f) Beam divergence (1/e diameter) is 0.30 mrad. With platform jitter, approximate effective divergence by

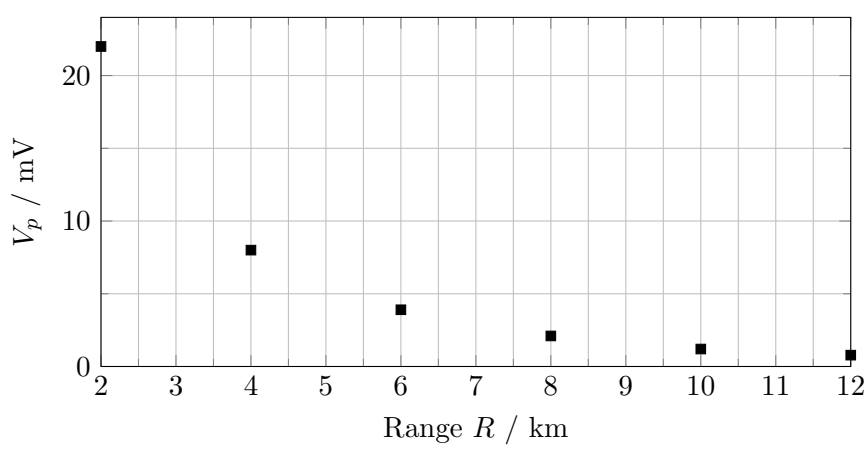
$$\theta_{\mathrm{eff}} \approx \sqrt{\theta_0^2 + (2\sigma_\theta)^2}.$$

Calculate θ_{eff} and the beam diameter at 6.0 km. [4]

(g) Returns from cloud droplets produce a strong reflection at 1.20 km. The true target is near 11.0 km.

Explain how range gating and coded pulse correlation together reject the cloud return while retaining the target return. [4]

(h) Fig. 4 shows measured received peak voltage V_p with range under constant alignment.



i. Test whether the data are more consistent with $V_p \propto 1/R^2$ or $V_p \propto 1/R^4$. Show brief working. [3]

ii. State one physical interpretation of the exponent you selected in (i). [1]

(i) During tracking, successive measured ranges are:

$$11\,040.0\,\mathrm{m},\ 11\,040.7\,\mathrm{m},\ 11\,041.5\,\mathrm{m},\ 11\,042.2\,\mathrm{m}$$

separated by one pulse period.

i. Estimate target radial speed. [2]

ii. Comment on whether the data suggest random noise only, or noise plus approximately uniform motion. [2]

[Total: 34 marks]

Suggested answers and mark scheme (Variant 2)

(a)

$$R = \frac{c\Delta t}{2} = \frac{(3.00 \times 10^8)(73.6 \times 10^{-6})}{2} = 1.104 \times 10^4\,\mathrm{m}$$

$R = 11.04\,\mathrm{km}$.
Marks: 1 formula, 1 value.

(b) i.

$$\Delta R = \frac{c\,\Delta t_u}{2} = \frac{(3.00 \times 10^8)(2.0 \times 10^{-9})}{2} = 0.30\,\mathrm{m}$$

Marks: 1 method, 1 value.

ii. Any one: faster timing clock/TDC, sharper pulse edge, improved signal-to-noise and fitting of pulse peak.
Marks: 1

(c) i.

$$E_p = P_{\mathrm{peak}}\tau = (1.6 \times 10^3)(8.0 \times 10^{-9}) = 1.28 \times 10^{-5}\,\mathrm{J}$$

$$P_{\mathrm{avg}} = E_p f = (1.28 \times 10^{-5})(5.0 \times 10^4) = 0.64\,\mathrm{W}$$

Marks: 1 for E_p, 1 for method for P_{avg}, 1 value.

ii.

$$E_\gamma = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{1.55 \times 10^{-6}} = 1.28 \times 10^{-19}\,\mathrm{J}$$

$$N = \frac{E_p}{E_\gamma} = \frac{1.28 \times 10^{-5}}{1.28 \times 10^{-19}} \approx 1.0 \times 10^{14}$$

Marks: 1 for photon energy, 1 for photon number.

(d)

$$T = \exp[-2(1.8 \times 10^{-4})(4.0 \times 10^3)] = e^{-1.44} \approx 0.237$$

Received energy:

$$E_r = E_p\,(3.0 \times 10^{-8})\,T = (1.28 \times 10^{-5})(3.0 \times 10^{-8})(0.237) \approx 9.1 \times 10^{-14}\,\mathrm{J}$$

Marks: 2 for transmission factor, 2 for received energy method/value.

(e) i. Set $V(t) = V_T$:

$$t = \tau_r \ln\left(\frac{V_0}{V_T}\right) = 12 \ln\left(\frac{42}{18}\right) \text{ ns} = 10.2\,\text{ns}$$

so $V > V_T$ for about 10 ns after the peak.
Marks: 1 equation use, 1 value.

ii. Higher threshold triggers later on rising edge (or may miss weak pulses), biasing time pick-off and causing systematic over/under-estimate of range.
Marks: 2 for clear causal explanation.

(f)

$$\theta_{\text{eff}} = \sqrt{(0.30 \times 10^{-3})^2 + (60 \times 10^{-6})^2} = 3.06 \times 10^{-4}\,\text{rad}$$

Beam diameter at 6.0 km:

$$d \approx \theta_{\text{eff}} R = (3.06 \times 10^{-4})(6.0 \times 10^3) = 1.84\,\text{m}$$

Marks: 2 for effective divergence, 2 for beam diameter.

(g) Expected points:

- Range gate opens only near expected round-trip time for 11 km, excluding early cloud echo at 1.2 km.
- Correlator checks correct pulse-position code over frame; cloud scatter is uncorrelated and rejected.
- Joint time-window + code requirement greatly lowers false-alarm probability.
- True target return satisfies both timing and coding constraints, so retained.

Marks: any 4 valid linked points.

(h) i. Check scaling:
For $1/R^2$, constant $k_2 = V_p R^2$: at 2 km, $k_2 = 22 \times 4 = 88$; at 10 km, $k_2 = 1.20 \times 100 = 120$ (not close).
For $1/R^4$, constant $k_4 = V_p R^4$: at 2 km, $k_4 = 22 \times 16 = 352$; at 10 km, $k_4 = 1.20 \times 10^4 = 12000$ (very poor).
Data are closer to $1/R^2$ than $1/R^4$ but with extra attenuation/system effects.
Marks: 2 for comparative test, 1 for conclusion.

ii. Interpretation: one-way inverse-square spreading dominates measured voltage trend; two-way and non-ideal receiver effects are partly compensated/calibrated in this measured peak quantity.
Marks: 1 for sensible interpretation.

(i) i. Pulse period $T = 1/f = 20\,\mu\text{s}$. Total range increase over 3 intervals:

$$11042.2 - 11040.0 = 2.2\,\text{m}$$

Time $= 3T = 60\,\mu\text{s}$, so

$$v \approx \frac{2.2}{60 \times 10^{-6}} = 3.7 \times 10^4\,\text{m}\,\text{s}^{-1}$$

Marks: 2 for method and value.

ii. Increments (0.7, 0.8, 0.7 m) are near-constant, indicating approximately uniform outward motion with small random noise.
Marks: 2

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Additional data for Variant 2:

- Laser wavelength $\lambda = 1.55\,\mu\text{m}$
- Pulse repetition frequency $f = 50\,\text{kHz}$
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- Peak optical power $P_{\text{peak}} = 1.6\,\text{kW}$
- Receiver lens diameter $D = 60\,\text{mm}$
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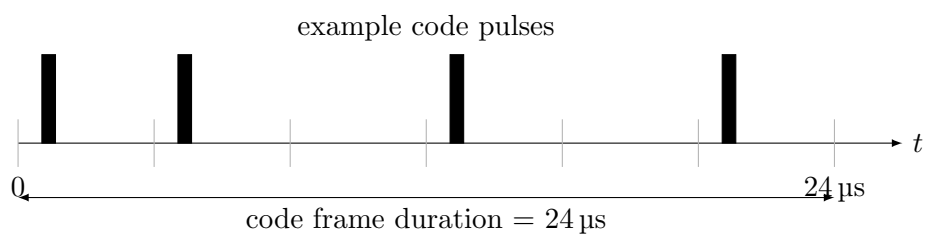

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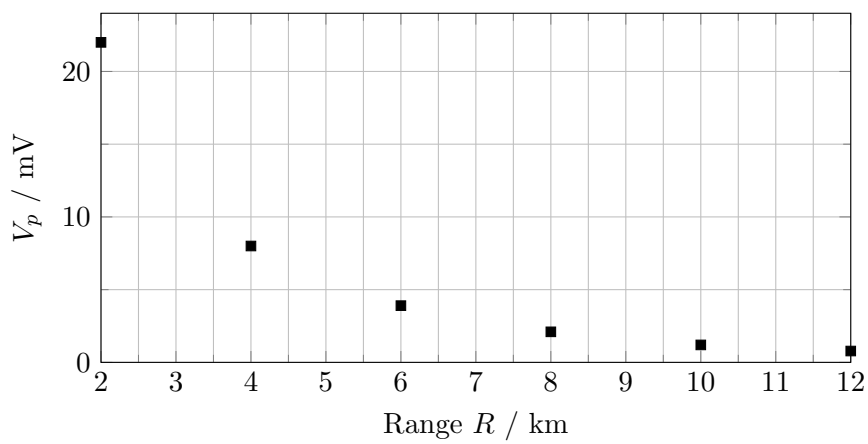

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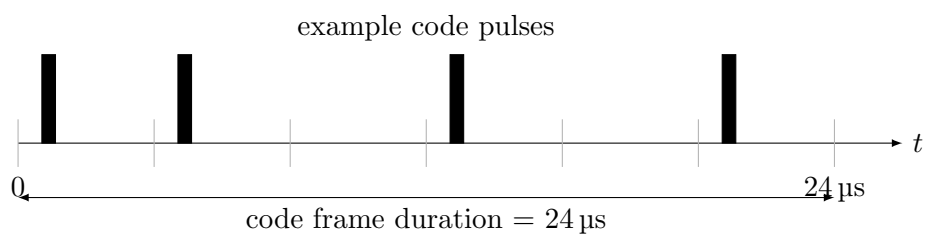
Fig. 3: Timing and coding frame



At one instant, the measured round-trip delay is $\Delta t = 73.6\,\mu\text{s}$ to a moving vehicle. The vehicle is known to move directly away from the instrument at approximately constant speed.

(a) Calculate the measured range at this instant. [2]

(b) The timing electronics have uncertainty of $\pm 2.0\,\text{ns}$ in locating pulse arrival.

i. Determine the corresponding uncertainty in range measurement. [2]

ii. Give one design change to reduce this timing-based uncertainty. [1]

(c) i. Calculate pulse energy and average transmitted power. [3]

ii. Calculate the number of photons emitted per pulse. [2]

(d) The round-trip atmospheric attenuation is represented by a transmission factor

$$T(R) = \exp(-2\alpha R), \quad \alpha = 1.8 \times 10^{-4}\,\mathrm{m}^{-1}.$$

For $R = 4.0\,\mathrm{km}$, calculate T and hence the received pulse energy if the overall geometric/reflectivity factor is 3.0×10^{-8} (before atmospheric attenuation). [4]

(e) The receiver transimpedance amplifier output for a detected pulse is approximated by

$$V(t) = V_0 e^{-t/\tau_r}, \quad \tau_r = 12\,\mathrm{ns},$$

where $t = 0$ at pulse peak and $V_0 = 42\,\mathrm{mV}$.

i. Determine the time interval after the peak for which $V(t) > V_T$. [2]

ii. Explain why excessive threshold voltage may cause systematic range error. [2]

(f) Beam divergence (1/e diameter) is 0.30 mrad. With platform jitter, approximate effective divergence by

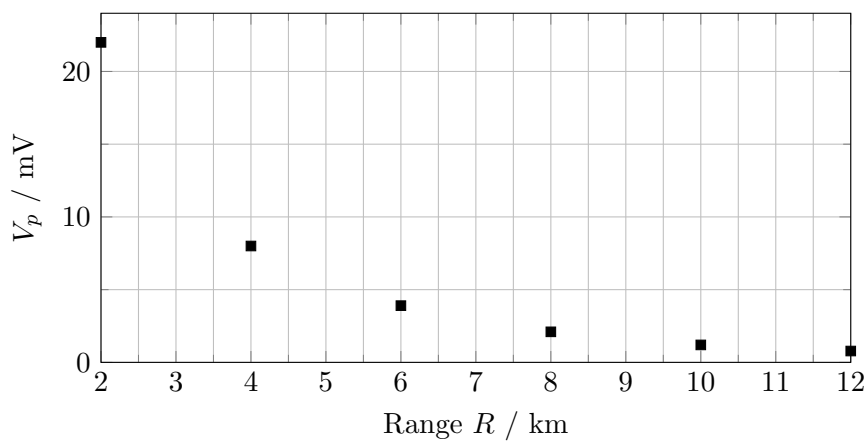
$$\theta_{\mathrm{eff}} \approx \sqrt{\theta_0^2 + (2\sigma_\theta)^2}.$$

Calculate θ_{eff} and the beam diameter at 6.0 km. [4]

(g) Returns from cloud droplets produce a strong reflection at 1.20 km. The true target is near 11.0 km.

Explain how range gating and coded pulse correlation together reject the cloud return while retaining the target return. [4]

(h) Fig. 4 shows measured received peak voltage V_p with range under constant alignment.



i. Test whether the data are more consistent with $V_p \propto 1/R^2$ or $V_p \propto 1/R^4$. Show brief working. [3]

ii. State one physical interpretation of the exponent you selected in (i). [1]

(i) During tracking, successive measured ranges are:

$$11\,040.0\,\mathrm{m},\ 11\,040.7\,\mathrm{m},\ 11\,041.5\,\mathrm{m},\ 11\,042.2\,\mathrm{m}$$

separated by one pulse period.

i. Estimate target radial speed. [2]

ii. Comment on whether the data suggest random noise only, or noise plus approximately uniform motion. [2]

[**Total: 34 marks**]

Suggested answers and mark scheme (Variant 2)

(a)

$$R = \frac{c\Delta t}{2} = \frac{(3.00 \times 10^8)(73.6 \times 10^{-6})}{2} = 1.104 \times 10^4\,\mathrm{m}$$

$R = 11.04\,\mathrm{km}$.
Marks: 1 formula, 1 value.

(b) i.

$$\Delta R = \frac{c\,\Delta t_u}{2} = \frac{(3.00 \times 10^8)(2.0 \times 10^{-9})}{2} = 0.30\,\mathrm{m}$$

Marks: 1 method, 1 value.

ii. Any one: faster timing clock/TDC, sharper pulse edge, improved signal-to-noise and fitting of pulse peak.
Marks: 1

(c) i.

$$E_p = P_{\mathrm{peak}}\tau = (1.6 \times 10^3)(8.0 \times 10^{-9}) = 1.28 \times 10^{-5}\,\mathrm{J}$$

$$P_{\mathrm{avg}} = E_p f = (1.28 \times 10^{-5})(5.0 \times 10^4) = 0.64\,\mathrm{W}$$

Marks: 1 for E_p, 1 for method for P_{avg}, 1 value.

ii.

$$E_\gamma = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{1.55 \times 10^{-6}} = 1.28 \times 10^{-19}\,\mathrm{J}$$

$$N = \frac{E_p}{E_\gamma} = \frac{1.28 \times 10^{-5}}{1.28 \times 10^{-19}} \approx 1.0 \times 10^{14}$$

Marks: 1 for photon energy, 1 for photon number.

(d)

$$T = \exp[-2(1.8 \times 10^{-4})(4.0 \times 10^3)] = e^{-1.44} \approx 0.237$$

Received energy:

$$E_r = E_p\,(3.0 \times 10^{-8})\,T = (1.28 \times 10^{-5})(3.0 \times 10^{-8})(0.237) \approx 9.1 \times 10^{-14}\,\mathrm{J}$$

Marks: 2 for transmission factor, 2 for received energy method/value.

(e) i. Set $V(t) = V_T$:

$$t = \tau_r \ln\left(\frac{V_0}{V_T}\right) = 12 \ln\left(\frac{42}{18}\right)\ \mathrm{ns} = 10.2\,\mathrm{ns}$$

so $V > V_T$ for about 10 ns after the peak.
Marks: 1 equation use, 1 value.

ii. Higher threshold triggers later on rising edge (or may miss weak pulses), biasing time pick-off and causing systematic over/under-estimate of range.
Marks: 2 for clear causal explanation.

(f)

$$\theta_{\text{eff}} = \sqrt{(0.30 \times 10^{-3})^2 + (60 \times 10^{-6})^2} = 3.06 \times 10^{-4}\,\text{rad}$$

Beam diameter at 6.0 km:

$$d \approx \theta_{\text{eff}} R = (3.06 \times 10^{-4})(6.0 \times 10^3) = 1.84\,\text{m}$$

Marks: 2 for effective divergence, 2 for beam diameter.

(g) Expected points:

- Range gate opens only near expected round-trip time for 11 km, excluding early cloud echo at 1.2 km.
- Correlator checks correct pulse-position code over frame; cloud scatter is uncorrelated and rejected.
- Joint time-window + code requirement greatly lowers false-alarm probability.
- True target return satisfies both timing and coding constraints, so retained.

Marks: any 4 valid linked points.

(h) i. Check scaling:
For $1/R^2$, constant $k_2 = V_p R^2$: at 2 km, $k_2 = 22 \times 4 = 88$; at 10 km, $k_2 = 1.20 \times 100 = 120$ (not close).
For $1/R^4$, constant $k_4 = V_p R^4$: at 2 km, $k_4 = 22 \times 16 = 352$; at 10 km, $k_4 = 1.20 \times 10^4 = 12000$ (very poor).
Data are closer to $1/R^2$ than $1/R^4$ but with extra attenuation/system effects.
Marks: 2 for comparative test, 1 for conclusion.

ii. Interpretation: one-way inverse-square spreading dominates measured voltage trend; two-way and non-ideal receiver effects are partly compensated/calibrated in this measured peak quantity.
Marks: 1 for sensible interpretation.

(i) i. Pulse period $T = 1/f = 20\,\mu\text{s}$. Total range increase over 3 intervals:

$$11042.2 - 11040.0 = 2.2\,\text{m}$$

Time $= 3T = 60\,\mu\text{s}$, so

$$v \approx \frac{2.2}{60 \times 10^{-6}} = 3.7 \times 10^4\,\text{m}\,\text{s}^{-1}$$

Marks: 2 for method and value.

ii. Increments (0.7, 0.8, 0.7 m) are near-constant, indicating approximately uniform outward motion with small random noise.
Marks: 2